

Soupal Ghosh

github.com/Soupal-Ghosh | [linkedin.com/in/soupalghosh](https://www.linkedin.com/in/soupalghosh) | soupal-ghosh.github.io/Portfolio
| soupalghosh15@gmail.com

SUMMARY

Computer Science (AIML) undergraduate specializing in deep learning, computer vision, and full-stack web development. Hands-on experience building production-ready AI systems using PyTorch, ResNet, and ArcFace for real-time face recognition, and deploying full-stack applications with Node.js and JavaScript. Completed DeepLearning.AI deep learning specialization with coursework in CNNs, hyperparameter tuning, and ML project structuring. Seeking ML/AI engineering or full-stack roles to apply research and development skills in real-world systems.

EDUCATION

2023 – 2027 **B.Tech in Computer Science & Engineering (AI/ML)** Institute of Engineering & Management, Kolkata

CGPA: 8.576/10

Relevant Coursework: Deep Learning, Database Management Systems (DBMS), Object-Oriented Programming (OOP)

PROJECTS

Aurawall — *HTML, CSS, JavaScript, Node.js, REST API*

[Live Demo](#)

- Engineered a full-stack wallpaper platform with an AI-powered image generation feature, allowing users to create and download custom wallpapers via a prompt-based interface.
- Architected a Node.js backend with REST API integration for AI image generation, paired with a fully responsive HTML/CSS/JavaScript frontend.
- Deployed on Render as a publicly accessible production web application.

Eye Disease Detection from Fundus Images — *Python, PyTorch, ResNet-50*

[Repository](#)

- Built a deep learning pipeline to classify retinal diseases from fundus images using a fine-tuned ResNet-50 model.
- Applied data augmentation and preprocessing techniques to improve model generalization across multi-class retinal conditions.
- Evaluated model performance using accuracy, validation loss, and confusion matrix metrics.

Voice-Aided Face Recognition System — *Python, OpenCV, ArcFace, OpenAI Whisper* [Repository](#)

- Designed a real-time face recognition system supporting live CCTV, recorded video, and webcam streams with voice-activated search.
- Leveraged SCRFD for face detection and ArcFace (ResNet-100) to generate 512-dimensional facial embeddings with cosine similarity matching for high-accuracy identity verification.
- Integrated dual speech recognition pipelines using Google Speech API and OpenAI Whisper for low-latency voice control.
- Research paper currently under preparation for publication.

- Building an analytics tool that fetches repository metadata via GitHub REST API to visualize commit activity, contributor statistics, and codebase metrics.
- Implementing automated data processing pipelines to generate statistical insights for repository health monitoring.

- Developing a convolutional autoencoder model to reconstruct degraded and occluded images by learning compressed latent space representations.
- Experimenting with denoising techniques and reconstruction loss functions to improve output image quality across varying noise conditions.

SKILLS

Languages	Python, Java, C, Basic JavaScript, HTML, CSS
ML / AI	PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, ResNet, ArcFace, NumPy, Pandas, Matplotlib
NLP / Speech	OpenAI Whisper, Google Speech API
Developer Tools	GitHub, VS Code, Eclipse, Render
Databases	MySQL
AI Tools	GitHub Copilot, Cursor, Claude, ChatGPT

CERTIFICATIONS

Python (Basic) Certification	HackerRank May 2026
SQL (Basic) Certification	HackerRank May 2026
Java (Basic) Certification	HackerRank May 2026
Problem Solving (Basic) Certification	HackerRank May 2026
Neural Networks and Deep Learning	DeepLearning.AI May 2025
Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization	DeepLearning.AI May 2025
Structuring Machine Learning Projects	DeepLearning.AI May 2025
Convolutional Neural Networks	DeepLearning.AI Dec 2025